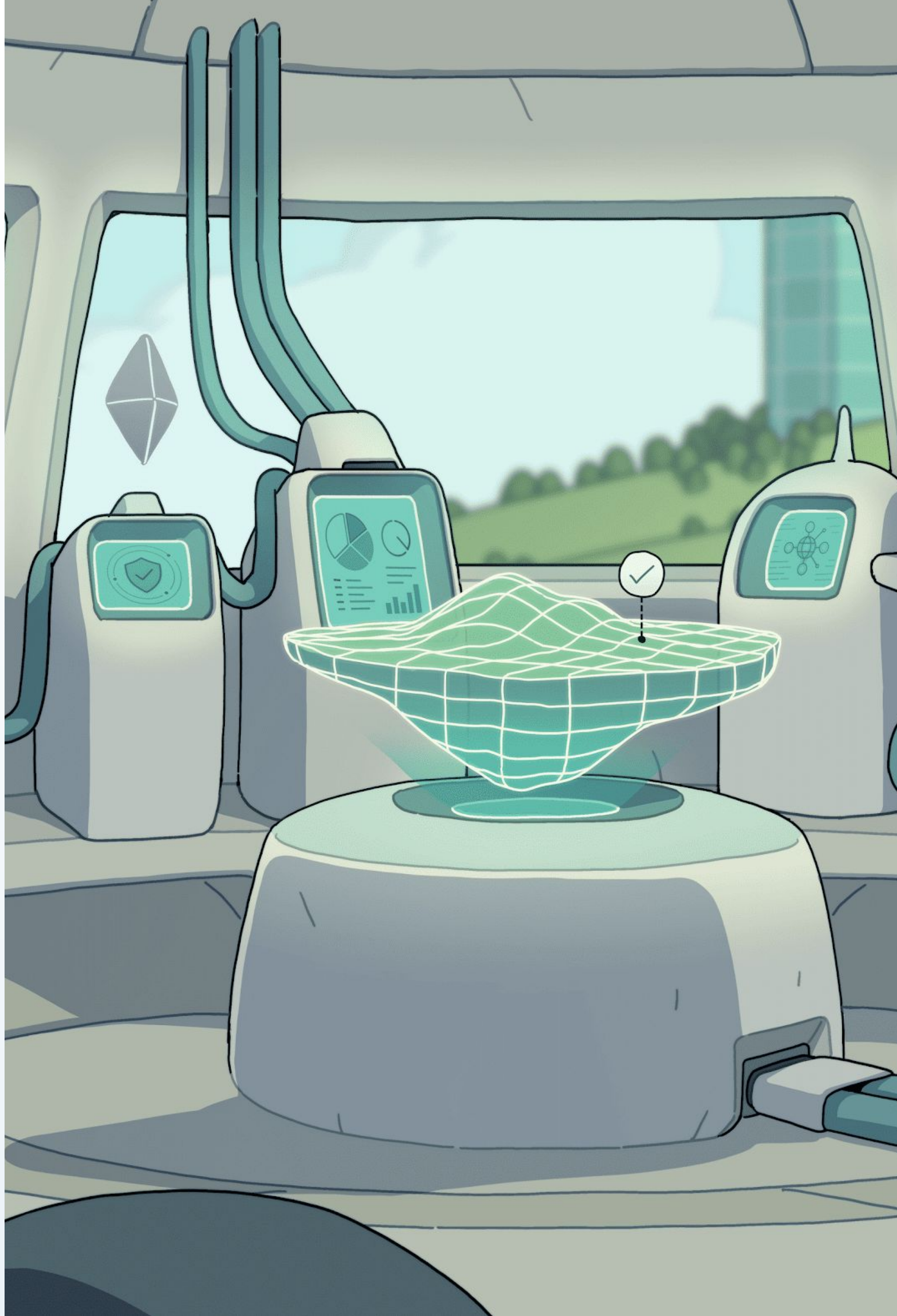
The road to AI in IT

Why the next generation of autonomous endpoint management requires infrastructure as code

July 2026



Executive summary

Infrastructure as code is the foundation for AI in IT

While 46.5% of IT leaders rank AI-driven automation as their highest priority for the year ahead, only 29.6% are investing in the infrastructure as code that makes it possible. This means seven out of ten organizations are trying to buy a finished outcome while completely skipping the baseline requirement.

AI ultimately runs on code. When device setups live in files that track changes over time, an engineer can simply describe a goal in plain language. The AI then writes the configuration, a human reviews and approves it, and the system logs the update so it can be undone in a single step. Software engineers have relied on this exact approach for a decade, safely using AI to write and alter their systems.

Enterprise device management has not reached that level of maturity. Currently, 87% of organizations operate without full automation, and 55% must juggle three or more separate tools to handle a single fleet. While companies like Stripe, Uber, Cursor, and Mollie already manage their devices as code, allowing AI to operate on their fleets today, teams relying on traditional management consoles cannot. This gap creates a sharp divide between the organizations putting AI to work on their devices and the ones falling further behind each quarter.

> Infrastructure as code is the defining moment between the IT teams putting AI to work on their fleets and the teams falling further behind each quarter. Without this foundation, AI automation remains completely out of reach.

Leaders are funding AI, not its foundation

46.5% are funding AI automation. Only 29.6% are investing in the infrastructure as code needed for full AI automation.

Shadow AI is a huge problem

The average enterprise runs 14 AI tools; IT can only see four. The risk lands on IT.

Very few organizations have autonomous fleets

46.5% rank AI a top priority. Just 13.5% run fully autonomous operations.

Same sh*t. Different day.

87% run multiple tools, 60% have unmanaged devices, 79% take more than a day to patch critical vulnerabilities.

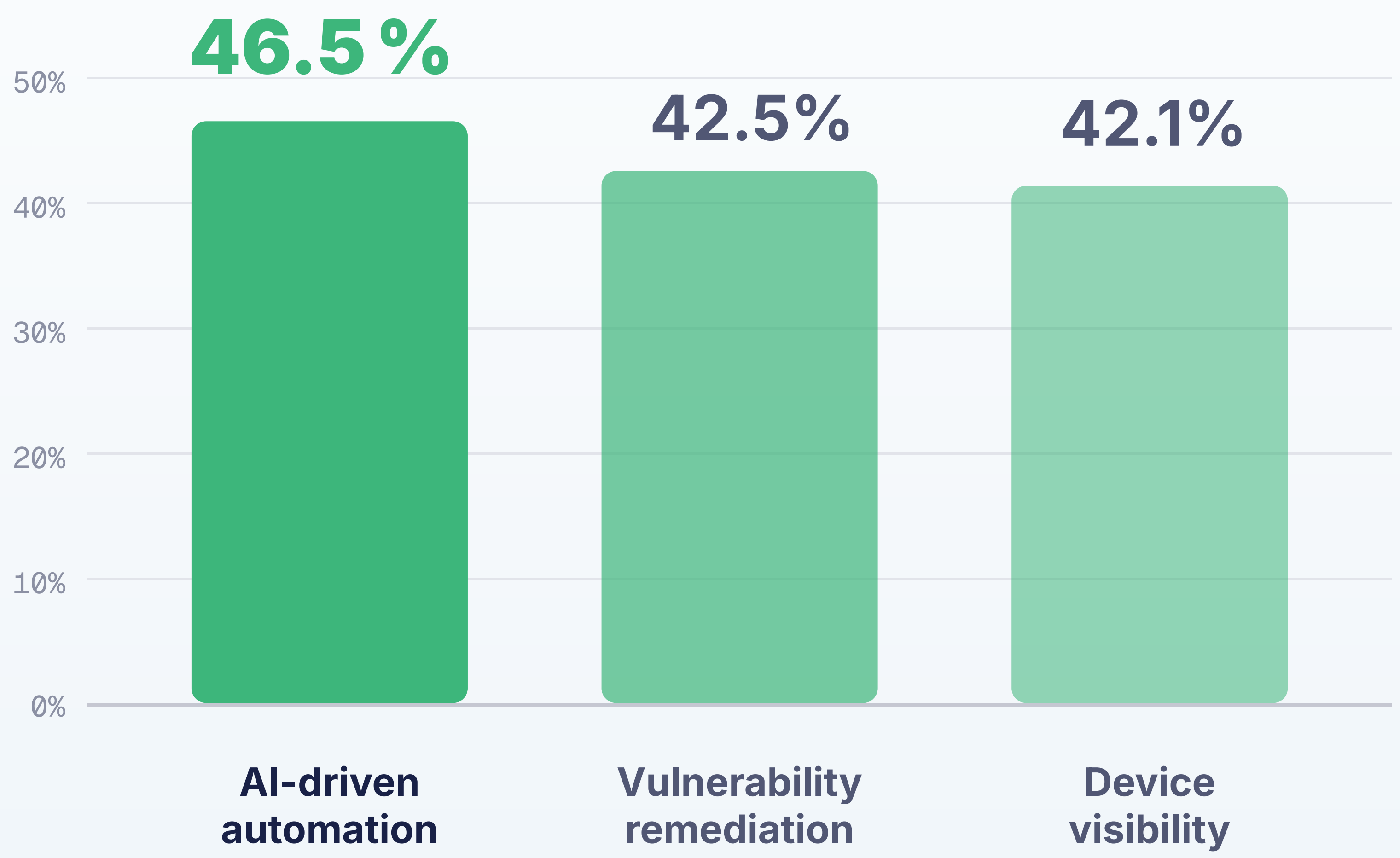
The teams already on infrastructure as code are ahead

What the teams already running on infrastructure as code do differently.

The industry-wide bet

Everyone is betting on AI in IT

Device automation leads all other categories in our survey, with 46.5% of IT leaders choosing it as a top investment priority for the coming year. Vulnerability remediation and device visibility followed closely behind at 42.5% and 42.1% respectively.



Share of 500+ leaders selecting each in their top three

The turn

Most organizations are skipping the foundation needed to deploy AI across their devices

70%

of [illegible]aders are not prioritizing the very th[illegible]equ[illegible]ed for AI automation.

46.5%

are investing in AI automation for device management

29.6%

prioritize infrastructure-as-code workflows

Top-three priorities · next 12 months · select up to three · n=500

The foundation

Why AI automation requires infrastructure as code

Traditional management consoles cannot provide the programmatic environment AI needs to read and modify configurations. Moving device setups into version-controlled files provides a declarative, GitOps-ready workspace where AI can analyze settings as structured configurations, propose precise changes via pull requests for human validation, and maintain an immutable deployment log to roll back any update instantly.

Readable

A readable surface

AI reads and writes structured config, not a screen it can't parse.

Reviewed

Human review

AI opens a pull request; an engineer reviews the diff and merges.

Reversible

Reversible by default

Every change is a commit, reversible to the last known-good state.

From the field

Enterprises managing their device fleets with infrastructure as code today

"Fleet allows the team to manage devices the same way they build software: through code, automation, and real-time data."

A global design platform

The browser-based tool millions of product teams use to design together

GitOps-first device management

~2,500 Macs, plus Linux & BYOD

"Compliance as code is why we chose Fleet. Access management, controls, least-privileged usage — all demonstrated in code from an endpoint perspective."

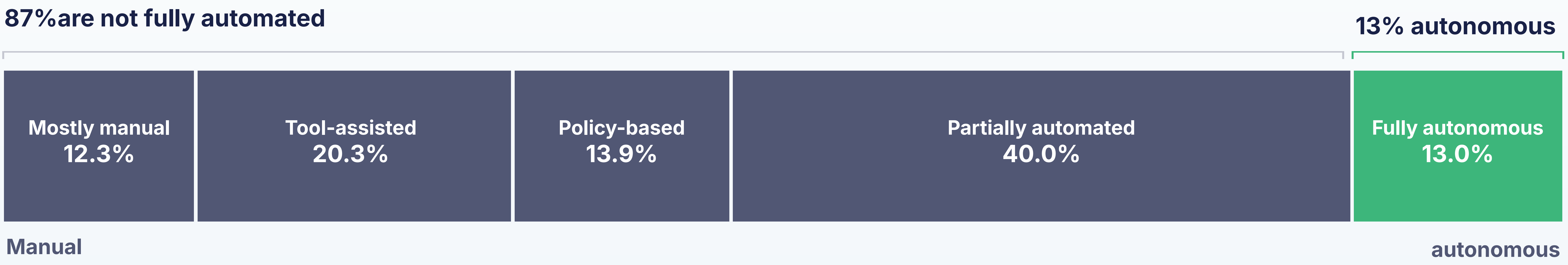
Sam Clarke
Senior Infrastructure Manager · Mollie

"My team is loving managing devices via GitOps, and the built-in CIS benchmark support makes it easy to enforce compliance across our devices."

Jeremy Baker
Engineering Manager · Faire

Automation maturity

Infrastructure as code is what makes automation possible, and only 13% have it



Manual, tool-assisted and policy-based

A third still execute tasks device by device, with no code to automate their device management.

The new baseline

40% are partially automated, but a human is still heavily involved.

Ready for autonomy

Only 13% have fully autonomous device management programs.

Shadow AI

Employees adopted AI. IT inherited the risk.

The support burden

IT secures the AI everyone else adopts, and gets none for its own work

01

Unsanctioned adoption is the norm

Engineers adopt AI tools without review. Token spend climbs, agents reach into production, and when something breaks the incident routes to IT.

02

No headcount to absorb it

No added headcount, the same fleet to manage, and a growing queue of support tickets for tools IT never sanctioned.

03

Locked out of the same tools

The teams adopting these tools have had AI in their workflows for years. The IT team supporting them still has none.

On the ground

"Shadow AI" is the industry's term for it. In practice it means more risk and more support load for IT, without the tooling to manage it.

Shadow AI by the numbers

AI is pervasive, and largely invisible to IT

78%

of employees use personal AI tools at work

Microsoft & LinkedIn Work Trend Index, 2024

14 AI tools (IT sees 4)

AI tools run in the average enterprise;
IT actually knows about only four of them

Productiv, 2026

The cost of ungoverned AI

Ungoverned AI spends serious money before anyone measures the return

Wasted token spend

4 months

for one company to burn through its entire 2026 AI budget after rolling an AI coding tool out to roughly 5,000 engineers.

Voss Solutions, 2026

+$670K

added to the average breach when shadow AI is involved: $4.63M against a $3.96M norm

IBM Cost of a Data Breach, 2025

Finance is already asking IT to account for AI spend. Most of it runs ungoverned and unmeasured.

Priority vs. reality

AI is the top priority. IT still isn't equipped to use it

The priority

46.5%

rank automating device management with AI as a top-three investment priority.

The reality

The teams asked to do more with less are the ones still locked out of AI.

Developers have had AI in their workflow for years. IT operations still runs the fleet by hand.

The payoff

The IT teams that do have AI in their workflow got there the same way developers did: by managing the fleet as code they can review.

Using vs. trusting

Teams won't automate changes they can't roll back themselves

87% have not delegated control to automated systems. Teams will not automate what they cannot review, roll back, or audit.

Infrastructure as code supplies the missing controls

Review
Every change is a pull request a person reviews before it merges.

Roll back
Version control means any change can be undone in one step.

Audit
Commit history is a record of who changed what, and when.

The known problems

Tool sprawl, blind spots, and slow patches

The problems every IT team already knows. Each one persists for the same reason: device management still lives in consoles, not code.

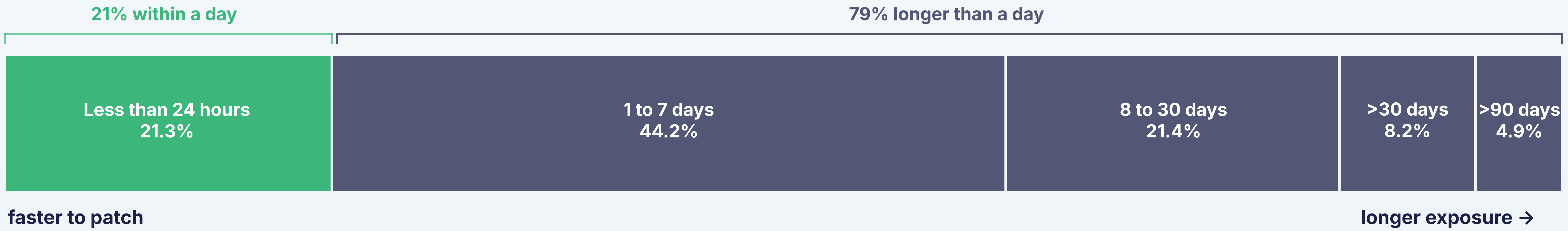
Patch velocity

79% take more than a day to patch a critical vulnerability

79%

take more than a day to deploy a critical security patch.

Modern exploits weaponize a vulnerability within hours; a remediation window measured in days is a standing exposure that only automated, code-driven remediation closes.



Tooling

Most run three or more tools to manage endpoints

MDM | PATCHING | INVENTORY | CONFIG | PROVISIONING | COMPLIANCE

87%

use two or more tools to manage and secure endpoints

55%

run three or more separate tools at once

13%

manage everything from a single platform

Device coverage

Six in ten have devices they can't fully manage

60% of organizations

The biggest blind spots cluster around mobile and Linux.

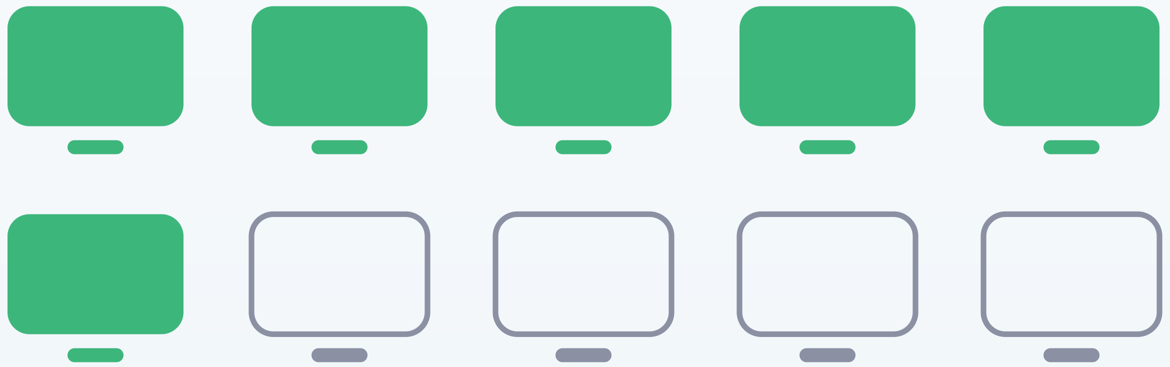
In practice

One communications platform found its previous tooling left coverage gaps across Linux servers and remote laptops. Consolidating onto a single, code-driven platform closed them.

Communications platform · ~3,000 endpoints

60%

of organizations



Each icon = 1 in 10 organizations

Onboarding

A productive first day is the exception

A new hire's first day should be zero-touch. For most organizations it is the same manual provisioning bottleneck that infrastructure as code eliminates.

59%

take more than 24 hours to deliver a new hire a fully provisioned device

28%

take four days or longer

A further **12% can't name a timeline at all,** so for 1 in 8, onboarding speed depends entirely on which device or department the hire lands in.

What to do now

Build on infrastructure as code now, or fall behind

The goal

Leaders agree on the goal – full visibility

Asked for their most important outcome over the coming year, leaders name one platform for every device. The goal is set; the infrastructure is the open question.

The goal needs a foundation

A single platform for every device only holds if those devices are managed as code. Without it, blind spots and configuration drift creep back in.

#1

The most-named one-year goal

Full visibility into every device from a single platform, is the #1 goal for organizations today.

The forward risk

The risks leaders fear are the ones infrastructure as code answers

The most common #1 three-year risk

Not patching critical vulnerabilities fast enough

Fixes ship as reviewed code

Remediation moves at the speed of a merge, not a ticket queue.

Keeping up with growth and device complexity	**One change, the whole fleet.** Code scales across every device without adding hands.
A disruptive change they can't roll back	**One-step rollback.** Every change is version-controlled and reversible by default.
Falling behind on AI adoption	**The on-ramp to AI.** IaC is the surface AI reads and writes. It is how AI reaches IT.

Single response · the four most-named #1 three-year risks, in order · n=500+

Proof in production

The teams already running on infrastructure as code are way ahead

10,000

Macs migrated to one platform in 10 days, saving $200K+ a year

Stripe

80% → <1%

new-hire setup error rate, cut from four in five to under one in a hundred

A top GPU manufacturer

$5–6M

saved on hardware by planning refreshes from live inventory data

Data platform · ~10,000 devices

15 min

to collect fleet telemetry that previously took 24 hours, a 96% reduction

Cloud data platform · tens of thousands of endpoints

Every one of these runs on infrastructure as code.

The move

Teams that adopt infrastructure as code get AI automation. The rest fall behind.

The organizations that adopt infrastructure as code now will run AI across their fleets. Those that delay fall further behind each quarter. Three moves get them there.

01

Put your devices in code

Move management off the GUI and into version-controlled configuration AI can read and write.

02

Review every change before it ships

The AI proposes changes. A person reviews and approves everyone before it reaches a device.

03

Let AI do the work in between

Drafting, patching, and routine changes run through AI, on a surface that stays auditable and reversible.

Methodology

Who we asked

A survey of over 500 IT leaders conducted in May 2026. All respondents hold Director-level or above titles and have direct responsibility for endpoint device management within their organizations.

Fieldwork: May 2026 Respondents: 500+
Company size: 2,000+ employees

500+

IT decision-makers, Director-level & above responsible for device management

4 regions

North America, the UK, Western Europe & Scandinavia

2000+

employees per company surveyed (minimum)

May 2026

fieldwork window

Respondent titles

Respondents were CIOs, CTOs, and CISOs, alongside VPs and Directors of IT and Heads of Endpoint Management.

Countries

They came from the United States, Canada, the United Kingdom, Western Europe, Sweden, and Norway.

Who runs on Fleet

Five of this year's eight hottest IPOs run on Fleet

From hyper growth startups to public enterprises, more than 1,300 organizations manage their devices with Fleet.

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